CHARCOAT

PASSIVE FIRE PROTECTION

CharCoat IC

Comments on SEC - MSS

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COMMENTS ON THE SAUDI ELECTRICITY COMPANY MATERIALS STANDARD SPECIFICATION No. 15-TMSS-07 Revision 0

INTRODUCTION

This document has been prepared in response to the invitation of the 5th January 2016 from the Saudi Electricity Company to comment on their Transmission Materials Standard Specification No. 15-TMSS-07. Being given the opportunity to do so is much appreciated.

Roy Macey, the author of this report, received his BSc(Eng) degree in 1972 and MSc(Eng) degree in 1981. He is a registered Professional Engineer, a Fellow of the South African Institute of Electrical Engineers, a Member of the Institute of Electrical and Electronic Engineers (USA) and a Distinguished Member of Cigre.

Mr Macey commenced his studies of insulator pollution at the University of Cape Town in 1971 with his undergraduate thesis on the subject. He continued his insulator investigations for a further four years with the national utility, Eskom's, Research Division. This was followed by twelve years as the specialist on overhead lines and insulators with the Cullinan Group. During this time, he designed all of the porcelain insulators manufactured in South Africa. In 1990 he founded Mace Electrical Technologies who, amongst other high voltage activities, developed the CharCoat IC silicone rubber insulator coating and the CharCoat IC insulator upgrading system.

He has delivered numerous papers around the world and presented courses on high voltage insulators in South Africa, Namibia, New Zealand, Australia, Norway and the United States. He is co-author of the book "The Practical Guide to Outdoor High Voltage Insulators". His paper "The performance of high voltage outdoor insulation in contaminated environments", presented to the SAIEE in 1980, gained a South African Transport Services Award. In 1987, he was awarded the Telemecanique Electrical Design Award and a Design Institute/Shell Design Award for his 22kV overhead line insulation system.

The comments provided in this document are based on the testing and overall service experience gained over the last thirty five years with the CharCoat IC silicone rubber coating material. The performance of CharCoat IC in some of the most severely polluted environments in the world is well proven. Tens of thousands of litres have been applied on line and substation insulation of system voltages up to 765kVAC and 533kVDC. However, being of a more advanced formulation than other insulator coatings on the market, its characteristics deviate from several clauses in the SEC specification. The difference in the properties and the resultant effects on field performance are described below.

DESIGN AND MATERIAL REQUIREMENTS

Clause 4.3.1, Table 01: RTV Coating Properties as Supplied.

- "Type – One-part, RTV"

CharCoat Passive Fire Protection Inc selected a two-part RTV silicone rubber for the CharCoat IC product. This means that a pre-measured small bottle of Catalyst is supplied with each container of CharCoat IC. The catalyst is poured into the 1 litre or 5 litre can of coating which is then shaken prior to use. (As the SEC Standard for the application of insulator coating, TCS-P-122.22, Clause 8.1.2(b), states that the "coating shall be mixed thoroughly to ensure even dispersion before spraying", the mixing in of the catalyst does not involve any additional activity.) This is a quick and simple operation and offers the following advantages:

  - o The curing of the rubber is more controlled and not dictated by the amount of moisture in the air.
  - o Because an outside agent is required for the rubber to cure, the shelf life of the material is considera bly extended.

Suggestion: Allow both one- and two-part coatings.

• **"Specific Gravity – 1.10 to 1.25"**
CharCoat IC is formulated with three different solvents, with different evaporation rates, for ease of application and to automatically yield a naturally uniform, smooth layer. This results in an "as supplied" SG of 1.0. Because the solvents evaporate, the amount has no effect on the subsequent performance of the coating and the SG is thus not a critical parameter.
Suggestion: If it is necessary to specify, allow coatings to have an SG from, 0.9 to 1.25.

• **"Skin-over time – 15 minutes"**
To have a skin-over time of 10 or 20 minutes should not cause any problem. To have to specifically formulate a coating to have a skin-over time of exactly 15 minutes at 25ºC ambient temperature and 50% relative humidity seems an unnecessary restriction.
Suggestion: Allow a skin-over time of, 10 to 20 minutes at the standard conditions.

• **"Percent Solids Contents, by weight - ≥ 70"**

The percentage solids by weight is a somewhat meaningless figure – it is dictated by the quantity of solvent present in the material. As the solvent evaporates, 100% solids are left on the insulator. It thus has absolutely no effect on the performance of the coating. Moreover, as Clause 4.4.6 states that "thinning/dilution shall be required" during application, then the percentage solids will be reduced from the "as supplied" value.

A far more important parameter is the ratio filler and other additives to the silicone content. The tendency is for manufacturers to include too much filler because it is significantly less expensive than the silicone and it improves the results of artificial tracking and erosion tests which intentionally destroy the hydrophobicity of the coating. Obviously, under service conditions, the longer hydrophobicity recovery and transfer times caused by lower silicone content, adversely affect the coating performance. In our opinion, the ratio of silicone rubber to other solids content should be at least 55:45, and ideally 60:40.
Suggestion: Rather specify a minimum silicone content of, say, 55% than a minimum solids content which just reduces the amount of solvent that can be used and necessitates the addition of more solvent during application.

• **"Dynamic Viscosity, cP – 1000 to 3500"**
The viscosity of CharCoat IC is 325cP. This is the reason that it is so easy to apply and yields a smooth, even finish which is superior to other products. When applying more viscous coatings, as stated in Clause 4.4.6, additional solvent needs to be added during the course of the process as the material has become too thick. CharCoat IC, with its lower viscosity, completely eliminates the need to handle additional hazardous materials during application.
Suggestion: A viscosity range of 200 to 2000cP would be more appropriate.

• Note 1: "The coating shall be supplied in White Color unless otherwise specified in the data schedule"
There is no evidence to show that a white coloured coating would perform any better than a grey or blue one.
Suggestion: Delete this note.

Clause 4.3.2, Table 02: RTV Coating Properties as Cured.
• "Dielectric Strength, V/mil (ASTM D149), minimum – 350 to 560"
As a minimum value is specified, a single figure, not a range, should be stated.
Suggestion: Specify a single value of, say, 500V/mil.

- **"Volume Resistivity, Ohm.cm (ASTM D257) – 1x1014 to 37x1014"**

Only a minimum value should be specified – there should not be an upper limit
Suggestion: Just specify, say, 1x1014 ohm.cm minimum.

- **"Tracking Wheel Withstand, hours, minimum – 1000"**

The test is undefined. Details of the procedure are required in order to perform the test. Parameters such as wetting times, drying times, time allowed for hydrophobicity recovery, etc., are critical to the outcome.
It is not possible to accurately simulate and, in particular, accelerate service conditions in a laboratory test. Silicone rubber, with its time-dependent hydrophobicity transfer and recovery characteristics, makes the undertaking of a meaningful artificial test even more difficult. For this reason, evaluations of material performance and life are far better performed in a severe, but natural, environment.
Suggestion: If such a test is to be included in the specification, it needs to be fully defined and its relevance to in-service performance confirmed.Testing under natural environmental conditions should be permitted as an alternative.

- **"Dry Arc Resistance, Track, Seconds – 180 to 200"**

As mentioned above, the test procedure needs to be defined. Further, only a minimum value should be given – there should not be an upper limit.
Suggestion: If such a test is to be included in the specification, it needs to be fully defined. Only a minimum value should be stated. Testing under natural environmental conditions should be permitted as an alternative.

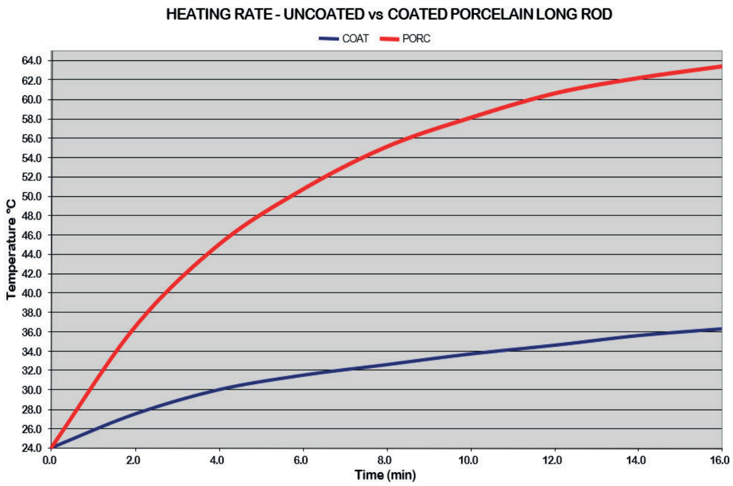
Clause 4.4.4.
It is not clear why the coating hardness should be specified. However, in general terms, the more elastic the coating is, the better. For example, wind borne sand will cause more abrasion of hard, rigid surfaces than flexible ones.
Suggestion: If it is considered that the hardness of the material needs to be specified, it should be a maximum, not a minimum, value.

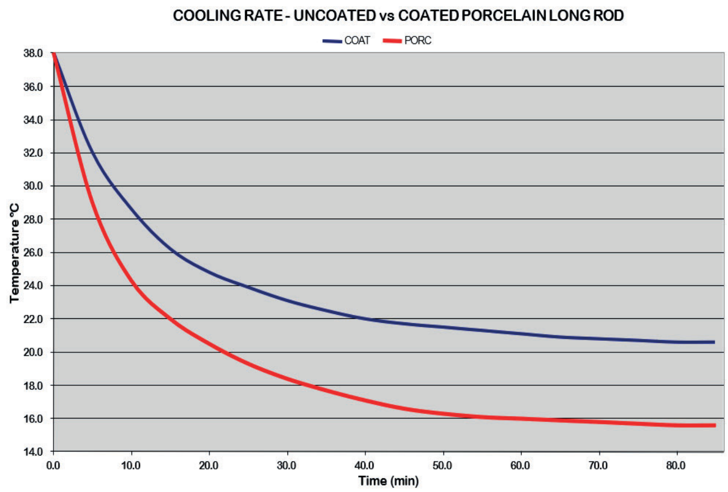
Clause 4.4.5.
CharCoat IC can be applied to any desired thickness but we consider 0,25 to 0,35mm to be the optimum range. Thermal transfer between the coating surface and the ceramic substrate should not be inhibited by too thick a layer.

The issue of coating thickness is addressed in Clause 6.6.2. of IEEE Standard 1523, which states:

"The normal thickness suggested by several manufacturers has been 0.5mm. This thickness is a practical "rule of thumb guide" that has been used for years in the coating industry and as such has little significance to coating of electrical insulators. The experience to date suggests that coating thickness is not a factor in either the performance or the life of the coating. Coating thicknesses in the range of 0.125 to 0.7mm have been applied in the field with equal success". "Laboratory tests indicate that when leakage current of a damaging magnitude develops on coatings, then thickness plays a role in the coating life. Thick coatings provide increased thermal resistance to heat generated by dry band arcing and as such do not allow the heat to be conducted away to the porcelain substrate as quickly as thinner coatings. Thick coatings can result in higher spot temperatures during dry band arcing, thereby causing thermal degradation of the coating sooner than thinner coatings".

The diagrams below show the insulator temperature change when coated and uncoated porcelain insulators are externally heated and cooled. This illustrates the high thermal resistance of silicone rubber and why thermal effects need to be considered.





Clause 4.4.6

As mentioned under Clause 4.3.1., Mace Technologies selected a two-part RTV silicone rubber for the CharCoat IC product. This means that a pre-measured small bottle of Catalyst is supplied with each container of CharCoat IC. The catalyst is added to the 1 litre or 5 litre can of coating which is then shaken prior to use. This is a quick and simple operation and offers the following advantages:

o The curing of the rubber is more controlled and not dictated by the amount of moisture in the air.

o Because an outside agent is required for the rubber to cure, the shelf life of the material is considerably extended.

The requirement that “thinning/dilution shall be required” and that the manufacturer must “supply the suitable thinner and submit a table showing the percentage quantity to be used” is astounding. Why this need for on-site hazardous material handling is condoned, whilst the addition of a small amount (180mm per 5 litres) of inert catalyst before spraying is forbidden, is incomprehensible. CharCoat IC requires no extra solvent addition and is thus safer in terms of both health and fire risks.

Suggestion: Allow both one- and two-part coatings. The need for the addition of hazardous materials during the application process should not “be required” but rather be forbidden.

Clause 4.4.8

Silicone rubber does not naturally adhere to other materials – hence its widespread use in the manufacture of moulds. Thus, if the application of a primer is prohibited, then a bonding agent has to be included in the coating itself. This means that the percentage of silicone in the coating is reduced and a significant volume of bonding agent is dispersed within the layer where it serves no beneficial purpose. With the separate application of a primer, the bonding agent exists only where it is required, i.e. at the interface between the coating and the porcelain, and no extraneous material is displacing the silicone content.

As illustrated below, after a 12 cycle,100 hour boiling test, the specimen with a pre-applied primer achieves far better adhesion than when 10% by weight of bonding agent is added to the coating or when no primer or bonding agent is used.

Suggestion: The use of a primer yields a higher silicone content coating and superior adhesion and thus should be permitted.

Clause 4.4.9.
As described above, excellent adhesion of silicone rubber to a ceramic substrate without a bonding agent contained within the material or applied as a primer, is not possible.
Suggestion: Remove the requirement "excellent unprimed adhesion".

Clause 4.4.10
For ease of use and to minimize wastage on site, CharCoat IC is supplied in 1 litre and 5 litre containers.
Suggestion: The supply of coating in containers of less than 20kg capacity should be permitted.

Clause 4.5.7
As mentioned for Clauses 4.3.1 and 4.4.6, we prefer a two-part silicone rubber to a one-part.

The evaporation rates of non-flammable solvents are generally too low for the easy and uniform spray application of silicone rubber coating. Further, the skin-over and tack-free times are increased. As most coating applications are undertaken under non-energized conditions, the flammability of the coating is not usually of concern.
Suggestion: Two-part coatings should be permitted. For better surface smoothness, uniformity and coverage, flammable solvents should be used. Non-flammable solvents should only be specified for specific live-line applications.

Clause 6.0
The values on the Data Schedule should be modified to be compatible with any changes made to the main body of the document.

GENERAL COMMENTS

The SEC Specification is written for the kind of coatings traditionally used by the utility. CharCoat IC, however, representing a more modern philosophy to the insulator pollution problem and thus of different formulation, is consequently not covered by the standard. Its effectiveness and superiority over other materials is, though, well proven. This is evident from Figure 2 which provides the result of coating evaluation tests at the severe natural test station at Eskom's Koeberg facility on the west coast of South Africa.

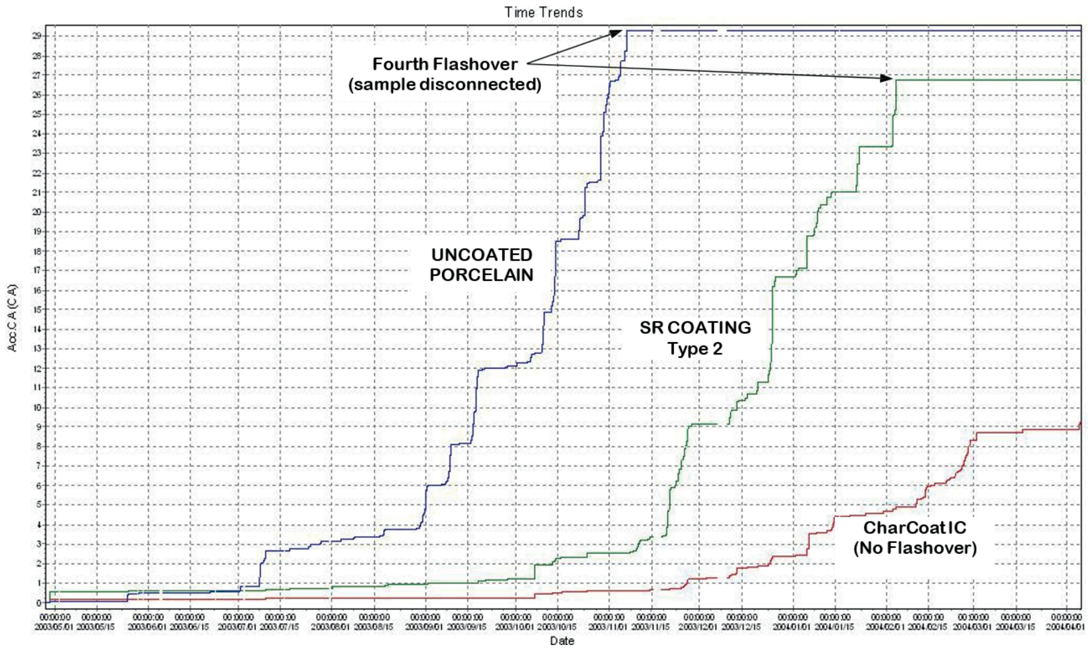


Figure 2: Accumulated coulomb ampere leakage currents measured on coated and uncoated porcelain insulators over a period of one year.

The graph indicates the accumulated leakage current activity over a one year period on three identical porcelain insulators – two of which were coated with different coatings and one left uncoated. Details of the test procedure and assessment method are defined in the Eskom Standard 34-224. In order for a product to pass the test and gain approval for use on the Eskom system, no more than three flashovers are permitted over the one year test period. As shown in the Figure, the uncoated porcelain had the highest leakage current activity and experienced its fourth flashover after 6 months. Coating Type 2 – a material used widely in the Middle East and complying fully with the SEC Specification – failed after suffering its fourth flashover in the ninth month. CharCoat IC displayed far lower leakage current amplitudes and had not one flashover in the year. CharCoat IC is the only coating to have achieved this distinction. (The Eskom Test Certificate is attached for reference).

Perhaps the main reason why CharCoat IC differs from its counterparts is the fact that it has not been developed by a polymer materials manufacturer as an additional outlet for their existing product range but by a company specializing in the design and supply of outdoor insulators and insulation upgrading solutions.